

Chapter 6

Introduction to Estimation

Learning Objectives

- Understand point and interval estimates and how it can be applied to engineering data.

6.0 Estimation

6.1 Introduction

The basic reasons for the need to estimate population parameters from sample information is that it is ordinarily too expensive or simply infeasible to enumerate complete populations to obtain the required information. The cost of complete censuses may be prohibitive in finite populations while complete enumerations are impossible in the case of infinite populations. Hence, estimation procedures are useful in providing the means of obtaining estimates of population parameters with desired degree of precision. We now consider *estimation*, the first of the two general areas of *statistical inference*. The second general area is *hypothesis testing* which will be examined later.

The subject of *estimation* is concerned with the methods by which population characteristics are measured from sample information. The objectives are to present:

1. properties for judging how well a given sample statistic estimates the parent population parameter.
2. several methods for estimating these parameters.

There are basically two types of estimation: *point estimation* and *interval estimation*. In *point estimation*, a single sample statistic, such as $\bar{X}$, s , or $\bar{p}$ is calculated from the sample to provide a best estimate of the true value of the corresponding population parameter such as μ , σ or p . Such a statistic is termed a *point estimator*.

The function or rule that is used to estimate the value of a parameter is called an *estimator*. An *estimate* is a particular value calculated from a particular sample of observations.

On the other hand, an *interval estimate* consists of two numerical values defining an interval which, with varying degrees of confidence, we feel includes the parameter being estimated.

6.2 Properties of a Point Estimator

Unbiasedness: If the expected value or mean of all possible values of a statistic over all possible samples is equal to the population parameter being estimated, the sample statistic is said to be unbiased. That is, if the expected value of an estimator is equal to the corresponding population parameter, the estimator is unbiased.

Example 66

The sample mean is an unbiased estimator of the population mean.

$$E(\bar{X}) = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \mu$$

Efficiency: The most efficient estimator among a group of unbiased estimators is the one with the smallest variance. This concept refers to the sampling variability of an estimator.

Example 67

Let $\hat{\theta}_1, \hat{\theta}_2$ and $\hat{\theta}_3$ be unbiased estimators for θ . Let $\hat{\theta}_4$ be a biased estimator among this group of estimators. Assume that $\text{Var}(\hat{\theta}_4) < \text{Var}(\hat{\theta}_2) < \text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_3)$. Which of these estimators is most efficient?

Consistency: An estimator is consistent if as the sample size increases, the probability increases that the estimator will approach the true value of the population parameter. Alternatively, an estimator is consistent if it satisfies the following conditions:

1. $\text{Var}(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$
2. $\hat{\theta}$ becomes unbiased as $n \rightarrow \infty$

6.3 Interval Estimation

For most practical purposes, it would not suffice to have merely a single value estimate of a population parameter. Any single point estimate will be either right or wrong. Therefore, instead of obtaining only a single estimate of a population parameter, it would certainly seem to be extremely useful and perhaps necessary to obtain two estimators, say $\bar{X}_1$ and $\bar{X}_2$, and say with some confidence that the interval between $\bar{X}_1$ and $\bar{X}_2$ includes the true mean μ .

Thus, an interval estimate of a population parameter is a statement of two values between which it is estimated that the parameter lies.

We shall be discussing the construction of confidence intervals as a means of interval estimation.

The confidence we have that a population parameter, θ , will fall within some confidence interval will equal $(1 - \alpha)$, where α is the probability that the interval does not contain θ (i.e. the probability α is an allowance for error).

To construct a 95% confidence interval $\Rightarrow \alpha = 0.05$. That is, the probability is 0.05 that the value θ will not lie within the interval.

Note that

$$\alpha + (\text{Confidence limit level}) = 1.0$$

The larger the confidence interval, the smaller the probability of error (α) for the interval estimator.

Confidence Interval For μ (σ known)

A confidence interval is constructed on the basis of sample information. It also depends on the size of α . Assume the population variance σ^2 is known and the population is normal, then the $100(1 - \alpha)\%$ percent C.I for μ is given by

$$\bar{X} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

Simply written as

$$\bar{X} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where $Z_{\alpha/2}$ is the Z value representing an area $\alpha/2$ to the right and left tails of the standard normal probability distribution.

Example 68

The yield of a chemical process is being studied. From previous experience yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 88.75, 90.8, 89.95, and 91.3. Find a 95% two-sided confidence interval on the true mean yield.

Solution

$$n=5, \sigma = 3, \bar{x} = 90.48, Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$90.48 - Z_{0.025} \left(\frac{3}{\sqrt{5}} \right) \leq \mu \leq 90.48 + Z_{0.025} \left(\frac{3}{\sqrt{5}} \right)$$

$$90.48 - 1.96(1.3416) \leq \mu \leq 90.48 + 1.96(1.3416)$$

$$87.8505 \leq \mu \leq 93.1095$$

Example 69

A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is normally distributed with $\sigma = 0.001$ millimeters. A random sample of 15 rings has a mean diameter of $\bar{x} = 74.036$ millimeters.

- Construct a 99% two-sided confidence interval on the mean piston ring diameter.
- Construct a 95% confidence interval on the mean piston ring diameter.

Solution

$$n=15, \sigma = 0.001, \bar{x} = 74.036, Z_{\alpha/2} = Z_{0.005} = 2.575, Z_{\alpha/2} = Z_{0.025} = 1.96$$

(a)

$$74.036 - Z_{0.005} \left(\frac{0.001}{\sqrt{15}} \right) \leq \mu \leq 74.036 + Z_{0.005} \left(\frac{0.001}{\sqrt{15}} \right)$$

$$74.036 - 2.575(0.000258) \leq \mu \leq 74.036 + 2.575(0.000258)$$

$$74.036 - 0.000665 \leq \mu \leq 74.036 + 0.000665$$

$$74.0353 \leq \mu \leq 74.0367$$

(b)

$$74.036 - Z_{0.025} \left(\frac{0.001}{\sqrt{15}} \right) \leq \mu \leq 74.036 + Z_{0.025} \left(\frac{0.001}{\sqrt{15}} \right)$$

$$74.036 - 0.00051 \leq \mu \leq 74.036 + 0.00051$$

$$74.0355 \leq \mu \leq 74.0365$$

Example 70

ASTM Standard E23 defines standard test methods for notched bar impact testing of metallic materials. The Charpy V-notch (CVN) technique measures impact energy and is often used to determine whether or not a material experiences a ductile-to-brittle transition with decreasing temperature. Ten measurements of impact energy (J) on specimens of A238 steel cut at 60C

are as follows: 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, and 64.3. Assume that impact energy is normally distributed with $\sigma = 1J$. We want to find a 95% CI for μ , the mean impact energy. The resulting 95% CI is

$$\begin{aligned}\bar{x} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) &\leq \mu \leq \bar{x} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \\ 64.46 - Z_{0.025} \left(\frac{1}{\sqrt{10}} \right) &\leq \mu \leq 64.46 + Z_{0.025} \left(\frac{1}{\sqrt{10}} \right) \\ 64.46 - 1.96 \left(\frac{1}{\sqrt{10}} \right) &\leq \mu \leq 64.46 + 1.96 \left(\frac{1}{\sqrt{10}} \right) \\ 64.46 - 0.6198 &\leq \mu \leq 64.46 + 0.6198 \\ 63.84 &\leq \mu \leq 65.08\end{aligned}$$

That is, based on the sample data, a range of highly plausible values for mean impact energy for A238 steel at 60°C is $63.84J \leq \mu \leq 65.08J$.

Exercises:

1. A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 20$.
 - (a) Find a 95% CI for μ when $n = 10$ and $\bar{x} = 1000$.
 - (b) Find a 95% CI for μ when $n = 25$ and $\bar{x} = 1000$.
 - (c) Find a 99% CI for μ when $n = 10$ and $\bar{x} = 1000$.
 - (d) Find a 99% CI for μ when $n = 25$ and $\bar{x} = 1000$.
2. A civil engineer is analyzing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000$ (psi)². A random sample of 12 specimens has a mean compressive strength of $\bar{x} = 3250$ psi.
 - (a) Construct a 95% two-sided confidence interval on mean compressive strength.
 - (b) Construct a 99% two-sided confidence interval on mean compressive strength. Compare the width of this confidence interval with the width of the one found in part (a).

Confidence Interval For μ (σ unknown/ $n \geq 30$)

In practice, the standard deviation of a population σ , is not likely to be known. When σ is unknown and n is 30 or more, we proceed as before and estimate σ with the sample standard deviation s . the resulting $1 - \alpha$ large sample confidence interval for μ becomes

$$\bar{X} - Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

Example 71

1. A sample of 40 ten-year-old girls gave a mean weight of 71.5 and standard deviation of 12 pounds respectively. Assuming normality, find the
 2. 90% confidence interval for μ .
 3. 95% confidence interval for μ
 4. 99% confidence interval for μ .
5. A hospital administrator took a sample of 45 overdue accounts from which he computed a mean of \$250 and a standard deviation of \$75. Assuming that the amounts of all overdue accounts are normally distributed. Find the
 6. 90% confidence interval for μ .

7. 95% confidence interval for μ .
8. The 99% confidence interval for μ .

Confidence Interval For μ (σ unknown/ $n < 30$)

When the σ is not known and the sample size is small, the procedure for interval estimation of population mean is based on a probability distribution known as the *student t-distribution*. When the population variance is unknown, and the sample size is small, the correct distribution for constructing a confidence interval for μ is the *t-distribution*. Here, an estimate s must be calculated from the sample to substitute for the unknown population standard deviation. The *t-distribution* is used such that

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

, where

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{(n - 1)}}$$

The *t-distribution* is based on the assumption that the population is normal. A $100(1 - \alpha)\%$ CI for the population mean, μ with the population normal and σ unknown is given by

$$\bar{x} - t_{\alpha/2, \nu} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\alpha/2, \nu} \left(\frac{s}{\sqrt{n}} \right)$$

where

$$\nu = n - 1$$

Notice that a requirement for the valid use of the *t-distribution* is that the sample must be drawn from a normal distribution.

Example 72

A sample of 25 ten-year-old boys yielded a mean weight and standard deviation of 73 and 10 pounds respectively. Assuming a normally distributed population, find 90, 95 and 99 percent confidence intervals for the mean of the population from which the sample came.

Solution

$$n = 25, \bar{x} = 73, s = 10$$

$$\bar{x} \pm t_{\alpha/2} \left(\frac{10}{\sqrt{25}} \right)$$

$$73 \pm t_{0.05}(2)$$

$$73 \pm 1.711(2)$$

$$(69.578, 76.422)$$

Summary:

Confidence Interval for μ

Population Mean μ	Confidence Interval
Sample size	
Large	
→ σ assumed known	$\bar{X} \pm Z_{\alpha/2} (\sigma/\sqrt{n})$
→ σ unknown (<i>estimated by s</i>)	$\bar{X} \pm Z_{\alpha/2} (s/\sqrt{n})$
Small	
→ σ assumed known	$\bar{X} \pm Z_{\alpha/2} (\sigma/\sqrt{n})$
→ σ unknown (<i>estimated by s</i>)	$\bar{X} \pm t_{\alpha/2} (s/\sqrt{n})$

6.3 Confidence Interval For A Population Proportion

It is often necessary to construct confidence intervals on a population proportion. For example, suppose that a random sample of size n has been taken from a large (possibly infinite) population and that X ($\leq n$) observations in this sample belong to a class of interest. Then $\bar{P} = X/n$ is a point estimator of the proportion of the population p that belongs to this class. Note that n and p are the parameters of a binomial distribution. Furthermore, we know that the sampling distribution of $\bar{P}$ is approximately normal with mean p and variance $p(1-p)/n$ if p is not too close to either 0 or 1 and if n is relatively large. Typically, to apply this approximation we require that np and $n(1-p)$ be greater than or equal to 5. We will make use of the normal approximation in this regard.

Definition: If n is large, the distribution of

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{\bar{P} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

is approximately standard normal.

The $100(1-\alpha)\%$ CI for p is then given by

$$\bar{P} - Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \leq p \leq \bar{P} + Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

Note that the quantity $\sqrt{p(1-p)/n}$ in the equation above is called the **standard error of the point estimator** $\bar{P}$. Unfortunately, the upper and lower limits of the confidence interval obtained from this equation contain the unknown parameter p . However, a satisfactory solution is to replace p by $\bar{P}$ in the standard error, which results in

$$\bar{P} - Z_{\alpha/2} \sqrt{\frac{\bar{P}(1-\bar{P})}{n}} \leq p \leq \bar{P} + Z_{\alpha/2} \sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$$

This procedure depends on the adequacy of the normal approximation to the binomial. To be reasonably conservative, this requires that np and $n(1-p)$ be greater than or equal to 5. In situations where this approximation is inappropriate, particularly in cases where n is small, other methods must be used.

Example 73

1. A manufacturer of electronic calculators is interested in estimating the fraction of defective units produced. A random sample of 800 calculators contains 10 defectives. Compute a 99% confidence interval on the fraction defective.

Solution

$$n=800, x=10, \bar{p} = x/n = 0.0125, n\bar{p} = 10, n(1 - \bar{p}) = 790$$

$$\bar{p} - Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \leq p \leq \bar{p} + Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$0.0125 - Z_{0.05} \sqrt{\frac{0.0125(1 - 0.0125)}{800}} \leq p \leq 0.0125 + Z_{\alpha/2} \sqrt{\frac{0.0125(1 - 0.0125)}{800}}$$

$$0.0125 \pm 2.575(0.003928)$$

$$0.0125 \pm 0.0101 = (0.0025, 0.0226)$$

1. Of 1000 randomly selected cases of lung cancer, 823 resulted in death within 10 years. Construct a 95% confidence interval on the death rate from lung cancer.

Solution

$$n=1000, x=823, \bar{p} = x/n = 0.823, n\bar{p} = 823, n(1 - \bar{p}) = 177$$

$$\bar{p} - Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \leq p \leq \bar{p} + Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$0.823 - Z_{0.05} \sqrt{\frac{0.823(1 - 0.823)}{1000}} \leq p \leq 0.823 + Z_{\alpha/2} \sqrt{\frac{0.823(1 - 0.823)}{1000}}$$

$$0.823 \pm 1.96(0.0121)$$

$$0.823 \pm 0.0237$$

$$(0.7993, 0.8467)$$

Exercise

- i. A random sample of 50 suspension helmets used by motorcycle riders and automobile race-car drivers was subjected to an impact test, and on 18 of these helmets some damage was observed. Find a 95% confidence interval on the true proportion of helmets of this type that would show damage from this test.
- ii. In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. Construct the 95% confidence interval for the proportion of bearings in the population that exceeds the roughness specification.
- iii. A survey was conducted to study the dental health practices and attitudes of a certain urban adult population. Of 300 adults interviewed, 123 said that they regularly had a dental checkup twice a year. Obtain a 95% CI for p , based on these data.

Confidence Interval for the Difference Between Two Population Means (Variances Known)

There are instances where we are interested in estimating the difference between two population means. Here, from each of the populations a sample is drawn and from the data of each, the sample means $\bar{x}_1$ and $\bar{x}_2$ respectively, are computed. The estimator $\bar{x}_1 - \bar{x}_2$ yields an unbiased estimate of $\mu_1 - \mu_2$, the difference between the population means.

The quantity

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a $N(0, 1)$ distribution.

The $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$\bar{X}_1 - \bar{X}_2 \pm Z_{\alpha/2} \left(\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$$

for large sample sizes n_1 and n_2 respectively.

Example 74

- i. Tensile strength tests were performed on two different grades of aluminum spars used in manufacturing the wing of a commercial transport aircraft. From past experience with the spar manufacturing process and the testing procedure, the standard deviations of tensile strengths are assumed to be known. The data obtained are as follows: $n_1 = 10, \bar{x}_1 = 87.6, \sigma_1 = 1$ $n_2 = 12, \bar{x}_2 = 84.5$, and $\sigma_2 = 1.5$. If μ_1 and μ_2 denote the true mean tensile strengths for the two grades of spars, find a 90% confidence interval on the difference in mean strength $\mu_1 - \mu_2$.
- ii. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1	Machine 2		
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Find a 95% confidence interval for the difference in means.

- iii. Two different formulations of an oxygenated motor fuel are being tested to study their road octane numbers. The variance of road octane number for formulation 1 is $\sigma_1^2 = 1.5$, and for formulation 2 it is $\sigma_2^2 = 1.2$. Two random samples of size $n_1 = 15$ and $n_2 = 20$ are tested, and the mean road octane numbers observed are $\bar{x}_1 = 89.6$ and $\bar{x}_2 = 92.5$. Assume normality. Construct a 95% confidence interval on the difference in mean road octane number.

Confidence Interval For The Difference Between Two Population Proportions

The magnitude of the difference between two population proportions is often of interest. An unbiased point estimator of the difference in two populations is provided by the difference in the sample proportions, $\bar{p}_1 - \bar{p}_2$. When n_1 and n_2 are large and the population proportions are not too close to 0 or 1, the central limit theorem applies

$$Z = \frac{\bar{P}_1 - \bar{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

where Z is a standard normal random variable and thus, normal distribution theory may be employed to obtain confidence intervals.

A $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is given by

$$[\bar{p}_1 - \bar{p}_2] - Z_{\alpha/2} \sqrt{\frac{\bar{p}_1(1 - \bar{p}_1)}{n_1} + \frac{\bar{p}_2(1 - \bar{p}_2)}{n_2}} \leq p_1 - p_2 \leq [\bar{p}_1 - \bar{p}_2] + Z_{\alpha/2} \sqrt{\frac{\bar{p}_1(1 - \bar{p}_1)}{n_1} + \frac{\bar{p}_2(1 - \bar{p}_2)}{n_2}}$$

Example 75

- i. Two different types of injection-molding machines are used to form plastic parts. A part is considered defective if it has excessive shrinkage or is discolored. Two random samples, each of size 300, are selected, and 15 defective parts are found in the sample from machine 1 while 8 defective parts are found in the sample from machine 2. Construct a 95% confidence interval on the difference in the two fractions defective.
- ii. Two hundred patients suffering from a certain disease were randomly divided into two equal groups. Of the first group, who received the standard treatment, 78 recovered within three days. Out of the other 100, who were treated by a new method, 90 recovered within three days. The physician wished to estimate the true difference in the proportions who would recover within three days. Find 95% CI for $p_1 - p_2$.
- iii. In a study designed to assess the side effects of two drugs, 50 animals were given Drug A, and 50 animals were given Drug B. Of the 50 receiving Drug A, 11 showed undesirable side effects, while 8 of those receiving Drug B reacted similarly. Find the 90 and 95 percent confidence intervals for $p_A - p_B$.